3	(a)	The current in the battery is the sum of the currents through P and Q. The p.d. across P and Q must be the same as they are in parallel circuit arrangement. From the graph, the value of p.d. of P for 0.15 A is 2.7 V. From the graph, the value of current of Q for 2.7 V is 0.090 A. Hence the total current is 0.15 + 0.09 = 0.24 A	1 1
	(b)(i)	P.d. across R is 4.5 – 2.7 = 1.8 V Resistance of R = 1.8 V / 0.24 A = 7.5 Ω	1 1
	(b)(ii)	Straight line passing through the points (0,0) and (1.5 V, 0.20 A)	1
	(c)	Resistance of P = V/I = 2.7 V / 0.15 A = 18 Ω Resistance of Q = 2.7 V / 0.09 A = 30 Ω Given that $R = \frac{\rho l}{A} \quad \rightarrow R = \frac{4\rho l}{\pi d^2}$ $l = \frac{R\pi d^2}{4\rho}$ $\frac{l_P}{l_Q} = \frac{R_P d_P^2}{R_Q d_Q^2} \quad \rightarrow \frac{l_P}{l_Q} = \frac{18}{30} \times \left(\frac{2}{1}\right)^2 = 2.4$	1 1 1
	(d)	The effective resistance of P and Q will be larger as Q stops conducting. The p.d. across P increases (due to the potential divider method), or the current through P has increased. From the graph, the ratio of current to voltage is smaller, hence the resistance of P increases.	1 1